

Name: _____

Date: _____



Year 8 Mathematics

Investigation 1, 2015

Topic – Surface Area of Prisms

Take-Home Section

Important Information:

Although the complete take-home component is not worth any marks, it is essential in preparation for the in-class validation component. Knowledge and skills gained will be extended in the validation component. This in-class validation will be completed under test conditions on the day in which this take-home component is due. The take home component may be used when completing the validation.

Date out:	Week _____ Date _____	Date Due:	Week _____ Date _____
Take home component weighting:	0% of the year	In-class component weighting:	7.5 % of the year

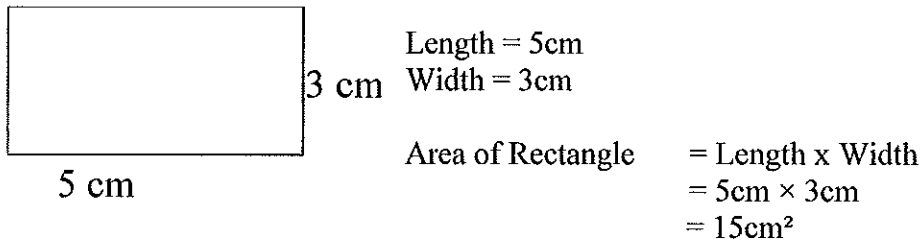
In this investigation you extend you knowledge and use of area of 2D shapes and apply them to simple Prisms

The first three pages are revision on Rectangles Parallelograms and Triangles which you should have covered in Year 7.

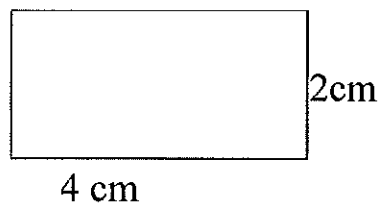
Then we will look at prisms and their nets.

Area of Rectangles

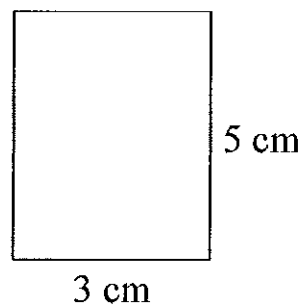
Here is how to set out an area of a rectangle question.



Use the above working out to calculate the area of these rectangles.



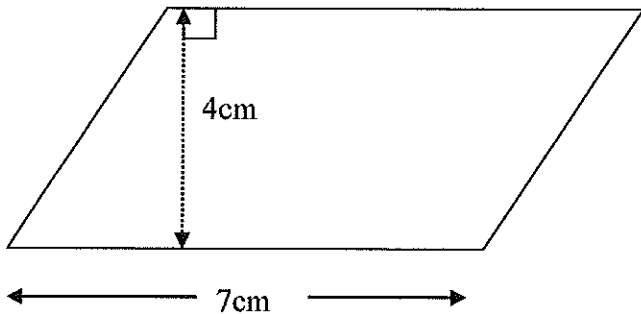
$$\begin{aligned} A &= L \times W \\ &= 4 \times 2 \\ &= 8\text{cm}^2 \end{aligned}$$



$$\begin{aligned} A &= L \times W \\ &= 5 \times 3 \\ &= 15\text{cm}^2 \end{aligned}$$

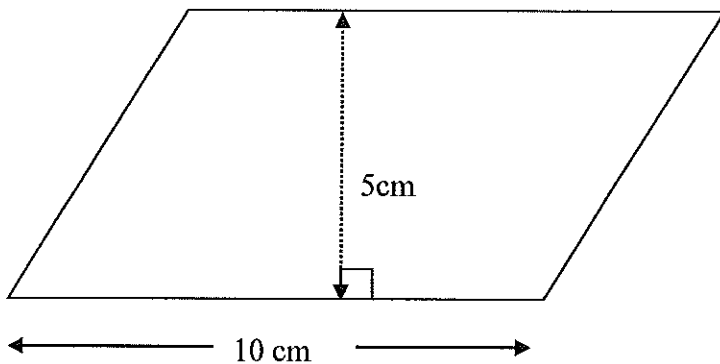
Area of Parallelograms

Area of a Parallelograms how to answer questions in full



$$\begin{aligned}\text{Area (Parallelogram)} &= PH \times B \\ &= 4\text{cm} \times 7\text{cm} \\ &= 28\text{ cm}^2\end{aligned}$$

Note the answer is in square units

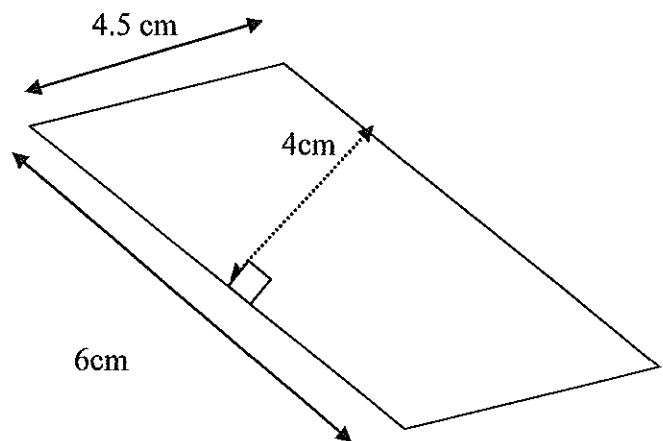


Show your full working out here

$$\begin{aligned}A &= PH \times B \\ &= 10 \times 5 \\ &= 50\text{ cm}^2\end{aligned}$$

Show your full working out here

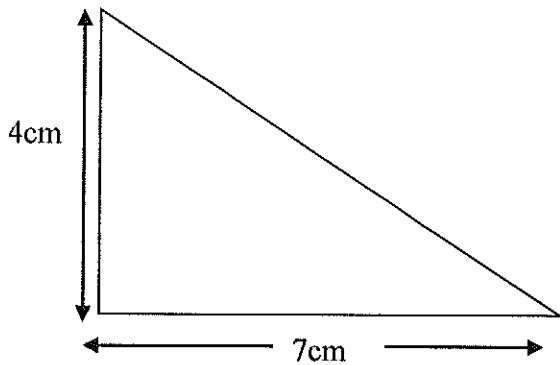
$$\begin{aligned}A &= PH \times B \\ &= 4 \times 6 \\ &= 24\text{ cm}^2\end{aligned}$$



Be careful which measurements you use.

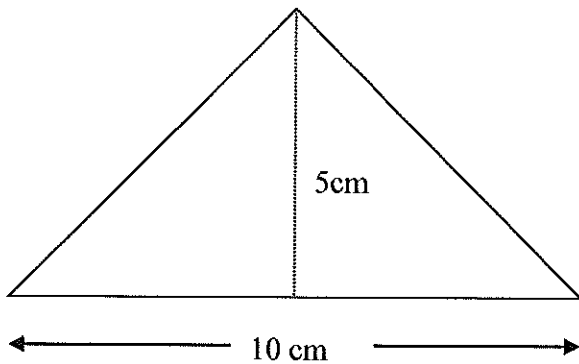
Area of Triangles

How to answer questions in full



$$\begin{aligned}\text{Area (Triangle)} &= \frac{1}{2} \times \text{PH} \times \text{B} \\ &= \frac{1}{2} \times 4\text{cm} \times 7\text{cm} \\ &= 14\text{ cm}^2\end{aligned}$$

Note the answer is in square units
PH stands for perpendicular Height

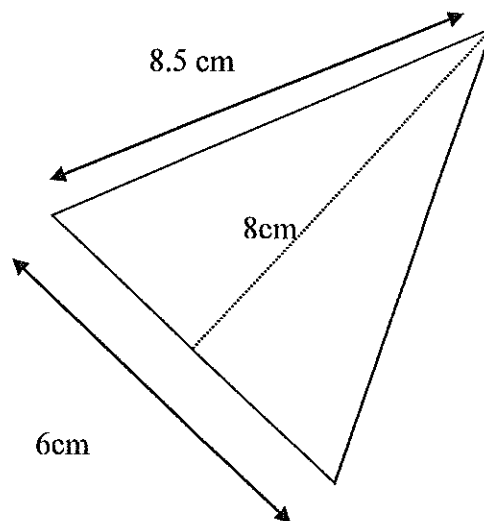


Show your full working out here

$$\begin{aligned}A &= \frac{1}{2} \times \text{PH} \times \text{B} \\ &= \frac{1}{2} \times 10 \times 5 \\ &= 25\text{ cm}^2\end{aligned}$$

Show your full working out here

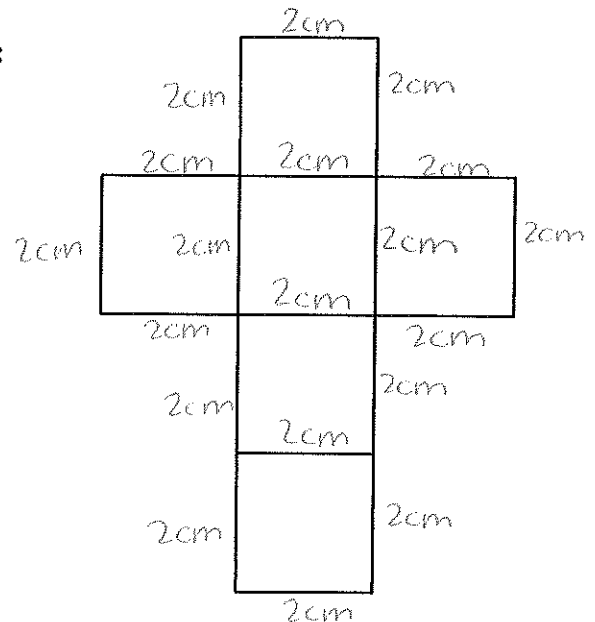
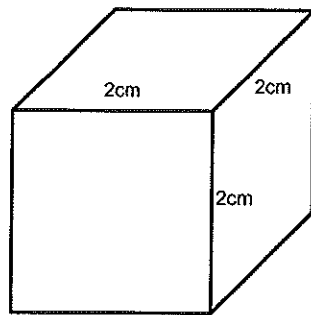
$$\begin{aligned}A &= \frac{1}{2} \times \text{PH} \times \text{B} \\ &= \frac{1}{2} \times 6 \times 8 \\ &= 24\text{ cm}^2\end{aligned}$$



Be careful which measurements you use.

Nets of shapes

Below is an isometric drawing of a cube with its net next to it:



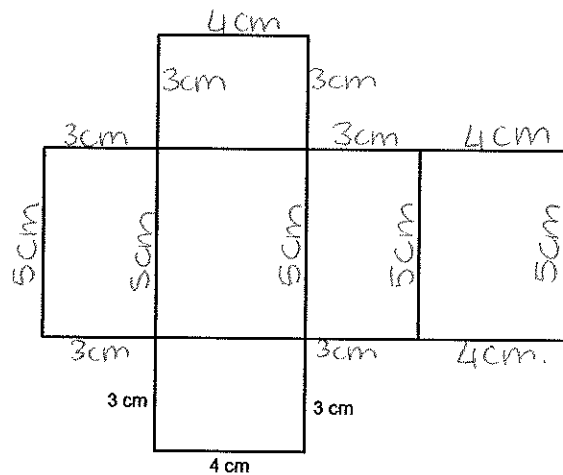
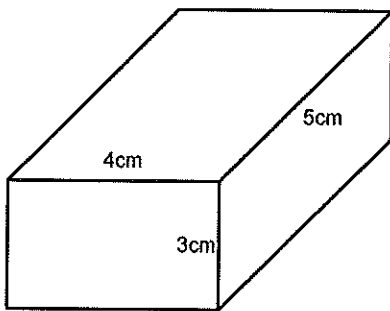
On the net label the length of each side.

What is the area of one side? $A = 2 \times 2 = 4 \text{ cm}^2$

What is the total surface area of the cube?

$$SA = 4 \times 6 = 24 \text{ cm}^2$$

Below is an isometric drawing of a rectangular prism with its net next to it:



There are 3 different rectangles in this net.

Label the length of each side some have already been done for you.

$$\begin{aligned} \text{Area(rectangle 1)} &= L \times W \\ &= 3 \times 4 \text{ cm} \\ &= 12 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area(rectangle 2)} &= L \times W \\ &= 3 \times 5 \\ &= 15 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Area(rectangle 3)} &= L \times W \\ &= 4 \times 5 \\ &= 20 \text{ cm}^2 \end{aligned}$$

$$\begin{aligned} \text{Total Surface Area} &= 2 \times \text{Area (Rectangle 1)} + 2 \times \text{Area (Rectangle 2)} + 2 \times \text{Area (Rectangle 3)} \\ &= 2 \times 12 + 2 \times 15 + 2 \times 20 \\ &= 94 \text{ cm}^2. \end{aligned}$$